Automation Of Driving License Tests Using Computer Vision and Image Processing

Prashanna Raj Pandit
Department of Electronics and Computer Engineering
Pulchowk Campus
Lalitpur, Nepal
prashannaraj5@gmail.com

Dewashish Atal
Department of Electronics and Computer Engineering
Pulchowk Campus
Lalitpur, Nepal
dewashishatal@gmail.com

Ayush Shakya
Department of Electronics and Computer Engineering
Pulchowk Campus
Lalitpur, Nepal
shakyayush2000@gmail.com

*Surendra Shrestha
Department of Electronics and Computer Engineering
Pulchowk Campus
Lalitpur, Nepal
surendra@ioe.edu.np

***Abstract*— Driving is a common and important mode of transportation in the daily lives of people. To drive legally a person requires a driving license, for which the candidate has to undergo a series of tests. This research paper explores an application of computer vision and image processing techniques in the context of automation of driving license tests. We propose CarSight, powered by advanced computer vision algorithms and image processing techniques designed to analyze visual data captured by strategically placed cameras during driving tests. By employing the YOLO-v5s framework, CarSight can accurately detect vehicles in real-time videos. The primary objective of CarSight is to provide an objective and automated evaluation of a driver's performance during the test. By analyzing various aspects such as adherence to specific rules, completion of the test within designated parameters, and overall driving proficiency, CarSight swiftly determines whether a candidate has passed or failed. This instant result ensures efficiency and transparency in the licensing process. It eliminates the subjective biases and errors associated with manual monitoring, ensuring fair and unbiased assessments. Additionally, it can mitigate the issue of bribery, especially in the context of Nepal by providing an objective evaluation that cannot be influenced or manipulated. The overall mAP (mean Average Precision) of the model CarSight was found to be 95%, indicating that the model is generally accurate in its predictions.**

***Keywords*— CarSight, YOLOv5, image processing, computer vision, driving license**

I. Introduction

A human vision system is so advanced that it can recognize any object very fast and accurately. For this reason, they are widely seen in security services for detecting anomalies. Similarly in the driving centers, especially in the context of Nepal, traffic authorities are present to watch the flaws encountered by the candidate. Since this system is totally manual, there are a lot of loopholes that result in fake license issues.

Fake licenses are issued mainly in two ways-one through "proxy" candidates and another by marking "pass" to the failed candidate. Some candidates have other people perform their trial examination; while many failed candidates bribe brokers to get licenses. Some such occurrences can be seen in the past. Police arrested four people for a fake driving license racket, where the group was found charging up to Rs.90,000 for providing a fake driving license (*https://kathmandupost.com/miscellaneous/2019/07/04/police-arrest-four-persons-of-a-fake-driving-licence-racket* (cited date: November 15, 2022)). Similarly, altering the test results have also seen in the past where an unsuccessful candidate was mentioned as 'pass' and the ones who successfully got through all the tests were stated as 'failed' ones and the ones who did not appear in the examination were also put in the group of 'pass' candidates (*https://www.peoplesreview.com.np/2022/03/18/ciaa-files-case-against-9-for-altering-driving-license-results/*(cited date: November 15, 2022)). These cases mostly occur with the joint agreement of traffic officials and brokers. As a result, people who hardly know how to ride the vehicles, get a license, resulting in accidents and large amounts of casualties.

Nepal contains various categories of driving licenses, ranging from A, B, to K for different types of vehicles. The sample track of the 4-wheeled vehicle is shown in Figure 1. This paper is aimed at real-time evaluation of driving license examinations using computer vision. The system detects and tracks the motion of 4-wheeler vehicles during the trial and evaluates the completion of the examination, according to government regulations.

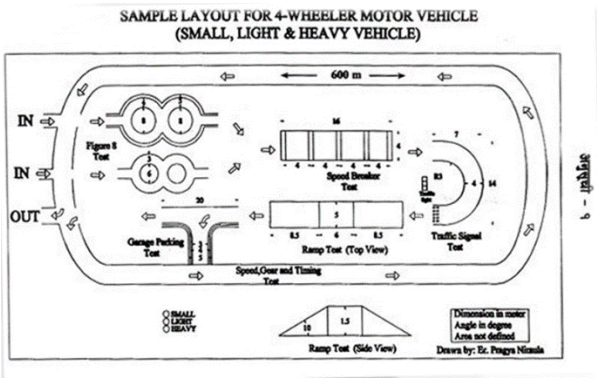


Figure 1. Sample Track for 4-wheeler Vehicle

(Source: Department of Transport Management)

*corresponding author
Surendra Shrestha
surendra@ioe.edu.np

The CarSight was deployed in our track layout, shown in Figure 2, and satisfactory results were observed. So, the implementation of CarSight in Nepal's driving license system could enhance fairness, transparency, and efficiency in licensing processes.

II. Related work

There have been many projects conducted worldwide in order to modernize the process of driving license trials and driving as a whole. An Arduino-based system with necessary sensor modules is developed for observing the candidate for getting their license. In this system, the candidates who take up the test were monitored and the result whether the candidate passed or failed was updated to the candidate as well as the authorities wirelessly using IOT and GSM modules [1]. An automated driving test using wireless sensor networks has also been developed in which the applicant is allotted the test vehicle for a test drive with the number of sensors embedded in the vehicle sending data using a wireless sensor network to a remote server to get processed. Result analysis is done by comparing the received data with previous data. This system is implemented using a Bayesian logic classification algorithm and feature extraction algorithm [2]. The technological solution for the trial center was also developed by customizing an 8051 controller-based embedded system and VB-based virtual instrument. This system used the controller module to sense the motion of the test vehicle on the test track and to evaluate the performance of the test driver based on the metric known as zero rpm measurement [3]. Similar research work and projects have been conducted to check the effectiveness of computer vision on driving performance. The proposed system uses computer vision and machine learning algorithms to detect and analyze the behavior of drivers during their driving license tests. The system was found to be effective in detecting unsafe driving behavior, such as speeding and reckless driving [4]. A system which includes a camera placed in the test vehicle to capture images of the driver and the road ahead, and utilizing features such as the driver's head orientation, eye gaze, and hand movement to evaluate the performance based on criteria such as adherence to traffic rules, safe driving behavior, and overall driving skills was found to be more effective than human examiners in detecting unsafe driving behaviors [5].
The ALT, a low cost smartphone based technology that automates key aspects of driving license tests has also been developed. The sole sensing platform is that of a windshield mounted mobile phone, where the front and rear cameras are used to monitor drivers' eyes as well as their inertial sensors for evaluation of driving maneuvers. For example, in order to ensure that the driver has scanned his mirror during a lane change, sensors are also used together [7].

III. Related Theory

You Only Look Once, or YOLO in short, is a state-of-the-art object detection algorithm capable of detecting objects via the neural network, with a single forward pass in real time [6]. YOLO-v5 is the fifth version of the algorithm that works by splitting an image into its grid and guessing the objects appearing in each grid cell. YOLOv5 is a one-stage object detection architecture based upon the modified version of EfficientNet's backbone network. YOLO-v5 models come in different sizes: YOLOv5s, YOLOv5m, YOLOv5l and YOLOv5x. YOLOv5s is the smallest version of YOLOv5 that contains 8 convolutional layers in the backbone network.

YOLOv5's architecture is mainly composed of three parts: backbone, neck and head. The backbone is designed using the New CSP-Darknet53 structure to partition the feature map of the base layer into two parts and then merges them through a cross-stage hierarchy. The neck is a combination of Spatial Pyramid Pooling Fast (SPPF) and New CSP-PAN structures. PANet is used to improve information flow and to localize pixels in the task of mask prediction; while SPPF is used in aggregation of the information received from the inputs and returning a fixed length output. The model head is composed of convolutional layers to predict locations of the bounding boxes, the scores and the object classes.

The ability to detect objects in a variety of scales and aspect ratios is one of the key features of YOLOv5. The network is predicting bounding boxes of objects that may differ in size and shape by using anchor boxes. Several other techniques that help improve the detection of objects are also included in this architecture, e.g. Feature Pyramid Networks (FPN) and cross-stage partial connections (CSP) which help to detect objects more accurately.
YOLOv5 employs various data augmentation techniques such as mosaic augmentation, copy-paste augmentation, MixUp augmentation, random affine transformation, albumentation and HSV augmentation to improve the model's ability to generalize and reduce overfitting. Various training strategies such as multiscale training, exponential moving average, AutoAnchor, warmup and cosine LR scheduler, mixed precision training and hyperparameter evaluation are implemented to enhance the model performance. The performance of YOLO models are analyzed by computing the combination of individual loss components: class loss–error in classification task, objectness loss–error in detecting presence of object in particular grid, and location loss–error in localizing object within grid cell [8].

IV. Methodology

A. Model Track Design

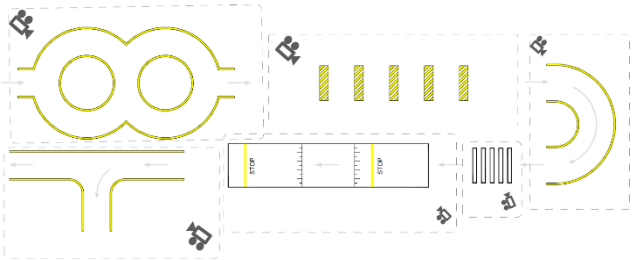


Figure 2. Model design of trial track (scale 1:13 in meter)

The model design of the trial center of Nepal was designed using AutoCAD is shown in Figure 2. The overall track is divided into six segments including the 8-test, speed breakers, U-turn, traffic lights and zebra crossing, ramp, and gar-

age parking. The individual segments of the track have a camera that records the vehicles in each segment. The 8-test is recorded by camera-1, similarly, the hump test, U-turn test, traffic light violation test, ramp test, and garage parking tests are recorded by camera-2, camera-3, camera-4, camera-5, and camera-6 respectively. These sections are to the scale of the remote-controlled car. All these six sections were then printed on a separate flex board, where the data collection and demonstration took place.

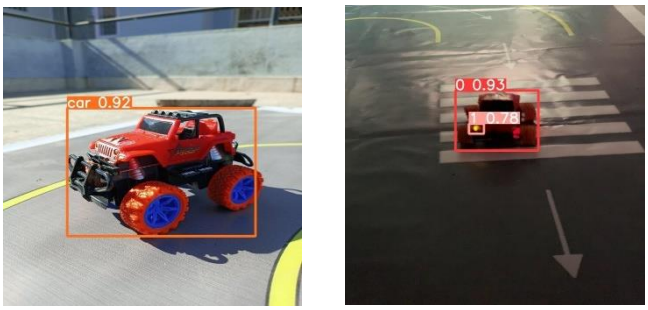
The scaling of the prototype is done with respect to the dimension of the remote controlled car. The actual width of the Maruti Alto which is provided as the trial vehicle is 1515mm. The width of the remote-controlled car that was used in the prototyping was 115mm. The defined ratio of the width of the trial car to the width of the track is 1:2. Hence the ratio of the width of the remote-controlled car to the width of the track in the prototype was also 1:2. So the ratio of the prototype track to the actual track was scaled to 1:13 in meter to meet all scaling and match the necessary dimensions.

B. *Making CarSight*

The training process involves collecting a dataset of approximately 200 images, capturing different parts of the trial process such as the 8-section, ramps, and U-turn. The dataset is annotated using a labeling tool, specifying bounding boxes for car and sidelight. The annotated dataset is then split into training, validation, and testing sets. The training set, comprising 70% of the data, is used to train the CarSight model and teach it how to perform. The validation set, which accounts for 20% of the data, helps fine-tune the model and prevent overfitting by monitoring its performance during training. The remaining 10% of the dataset is used as the testing set to evaluate the model's accuracy, precision, and overall performance.

The annotated data was converted to the required YOLO format, and the YOLOv5s training script was used for training. The Torch library and necessary modules were installed. The model was trained with a batch size of 16 and 150 epochs. Evaluation of the validation set was performed using the YOLOv5 evaluation script to assess the model's accuracy and identify areas for improvement.

The training results obtained are shown in Figure 5, and some outputs while testing CarSight are shown in Figure 3.



(a)Detection of car. (b) Detection of car and sidelight.

Figure 3. Bounding Box Prediction of Model Car: (a) detection of the car with 0.92 confidence score, (b) detection of car and side light with 0.93 and 0.78 confidence scores respectively.

The performance of the trained model was deemed to be satisfactory, so a YOLOv5 inference script was used to deploy the model and perform object detection on live video streams. The overall process of training the model is shown in Figure 4.

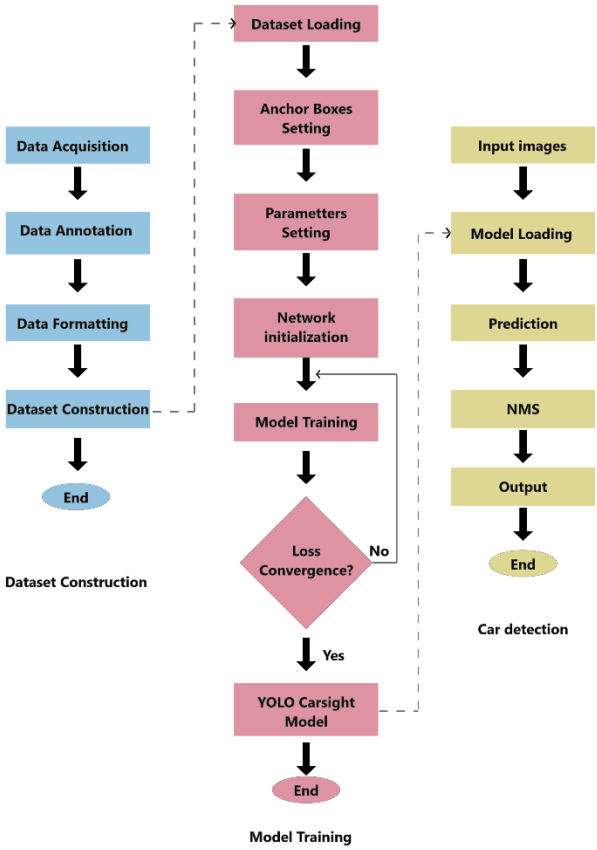


Figure 4. YOLO Carsight model flowchart for dataset, training, and detection process

C. *Load CarSight For real-time object detection*

We incorporated CarSight, the trained YOLOv5 model, into our main program where various decision-making tasks were performed. We read the input frames from the cameras that we have installed at 6 different locations. This was done by using the OpenCV library. Preprocessing of the input image must be done before running the model. The data obtained from the cameras may not all be uniform. So, all the images was resized to a consistent size. Video frames need to be preprocessed so that objects can be easily detected by the model. The opacity of the pictures may vary, due to differing surrounding lighting. Some images appear brighter, while others appear dark. So we preprocess the input frames by resizing, normalizing, and converting them to grayscale and apply Gaussian blur followed by binary thresholding.

The model was then run on each preprocessed image frame to get the predictions for bounding boxes, class identification, and confidence scores for each object in the frame. Post-processing steps such as non-maximum suppression were performed to remove overlapping bounding boxes and keep only the most confident ones. We put a confidence level of 40% and overlapping 30%. After getting the coordinates of the bounding boxes, the bounding box of the object was plotted. To find the intersection of the bounding box and track, the input image was converted to HSV Color space because of

the ease of segmenting color in the HSV Color space. The lower and upper threshold values of the yellow color track were noted as:

lower = np.array([0,179,171])

upper = np.array([255,255,255])

The binary mask of track was created within the specified range by inRange() which separates objects of interest from the background of an image. Now to find the intersection between the track and the bounding box, we also create the binary mask of the bounding box.

The intersection was obtained by bitwise AND operation between two images (i.e. track mask and bounding box mask). Now the number of non-zero pixels in the resulting image was counted by the 'countNonZero()' function. If the number of white pixels is greater than 100 then the alarm rings to declare the fail condition. The side light is also continuously monitored in the specific section.

In the traffic-light violation test, the pixel intensity of lights is monitored continuously. If the red pixel is greater than or equals to 200 then the red light is considered as ON. Here, an imaginary line is drawn on frames just ahead of zebra-crossing. When the car tries to cross the zebra crossing at a red light then the mid-point of the bounding box intersects with the line and hence generates the alarm.

One condition that universally results in failure is if the car turns off during any of the track segments. To detect this, the system counts the number of still frames, and if the counter reaches 150, the car is considered to be in a stopped position and the candidate fails. In other words, if the car remains stationary for more than 150 frames (i.e. 5 seconds if the video is 30fps), it is deemed a failed condition.

.

V. RESULTS AND DISCUSSION

The overall mAP(mean Average Precision), precision and recall of the model CarSight is 95%, 93.3% and 93.8% respectively. Figure 5 contains the precision, recall, and mAP of the two classes. Here, class 1 represents the sidelight, and class 0 & car both represent the car. The overall performance of the model is shown in Figure 6.

```
Model summary: 157 layers, 7018216 parameters, 0 gradients, 15.8 GFLOPs
                 Class     Images  Instances          P          R      mAP50   mAP50-95:
                   all         35         51      0.933      0.938       0.95      0.677
                     0         35         24      0.981          1      0.995      0.774
                     1         35         16      0.858      0.812      0.859      0.314
                   car         35         11      0.959          1      0.995      0.944
```

Figure 5. Model output

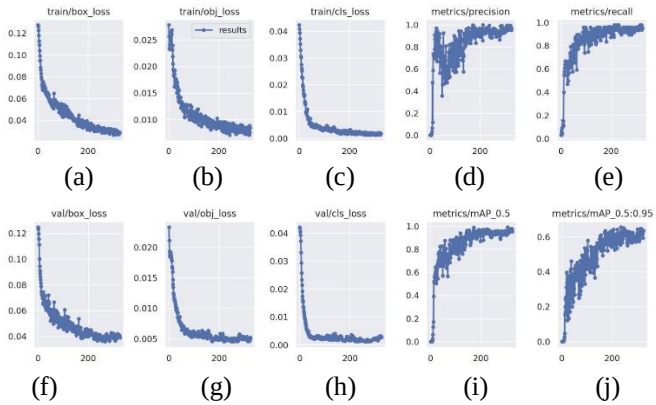


Figure 6. Training Results Output: (a) plot of box loss while training, (b) plot of object loss while training, (c) plot of class loss while training, (d) plot of precision, (e) plot of recall, (f) plot of box loss while validation, (g) plot of object loss while validation, (h) plot of class loss while validation, (i) plot of mAP at 0.5 IOU threshold, (j) plot of mAP calculated over a range of IoU thresholds from 0.5 to 0.95.

The F1 confidence curve, shown in Figure 7, begins at a 0 F1 score and increases with confidence, indicating improved accuracy. A significant improvement in performance is observed at a confidence level of 0.691, where the F1 score rises sharply.

The precision-confidence curve shown in Figure 8, shows an increasing trend. As the confidence level increases, the precision improves, reaching a peak precision of 1 at a confidence level of 0.945. This indicates that the model's predictions become highly accurate and confident.

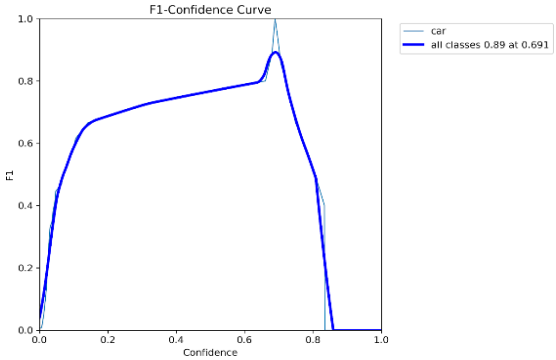


Figure 7. F1 Confidence Curve (%)

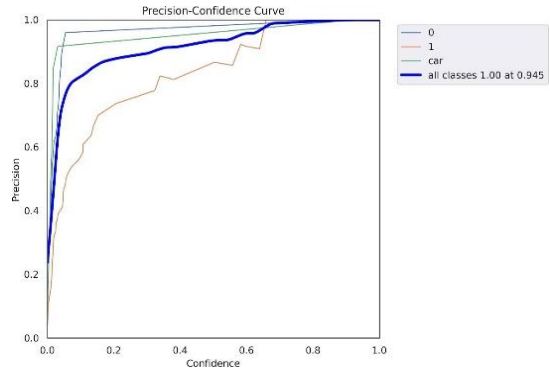


Figure 8. Precision Confidence Curve (%)

The precision-recall curve in Figure 9, shows that the model achieves high confidence in predicting the class "car" (0.995) and slightly lower confidence for the class "sidelight" (0.859). Overall, the model demonstrates an average precision of 0.95 mAP@0.5, indicating general accuracy in its predictions.

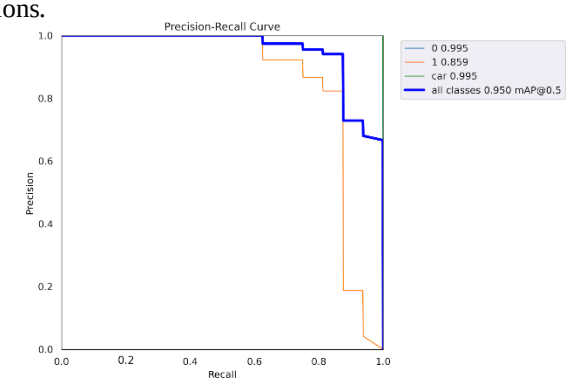
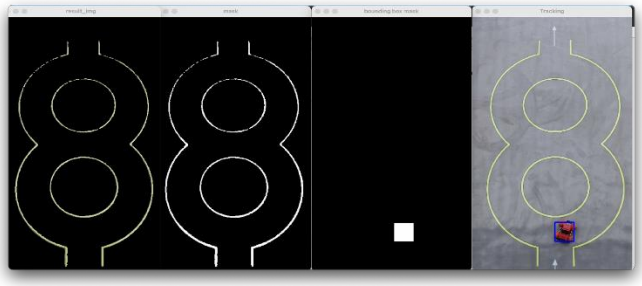


Figure 9. Precision-Recall Curve (%)

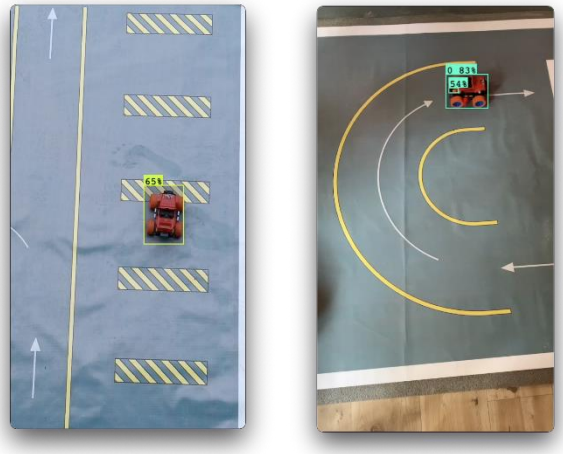
The output obtained after the implementation of CarSight in 8-test, hump, U-turn, traffic light, and parking are shown in Figure 10 and Figure 11. During the 8-test, as shown in Figure 10, the alarm is generated when the car touches the line by finding the intersection between the windows Bounding_box_mask and mask. Also when the class 0 (i.e. side-light) is detected.

The output of the hump track with its bounding box prediction is shown in Figure 11 (a). This section, output of camera-2, detects if the car travels in a straight line or not. If the car goes out of track or detects any side light blinking then the candidate is declared as fail. Figure 11 (b) shows the U-turn test which is taken from camera-3. The section checks for both line crossing and side lights. If the line is crossed or a side light is detected in this video frame then the candidate is declared as fail. Similarly, Figure 11 (c) shows the traffic light violation test. This frame is taken from camera-4, and Figure 11 (d) shows the output of the garage parking test, taken from camera-5. The image shows a car on a parking track with bounding box prediction. The candidate is tracked along their path, and an alarm is sounded to denote failure when the vehicle touches the yellow boundary line.

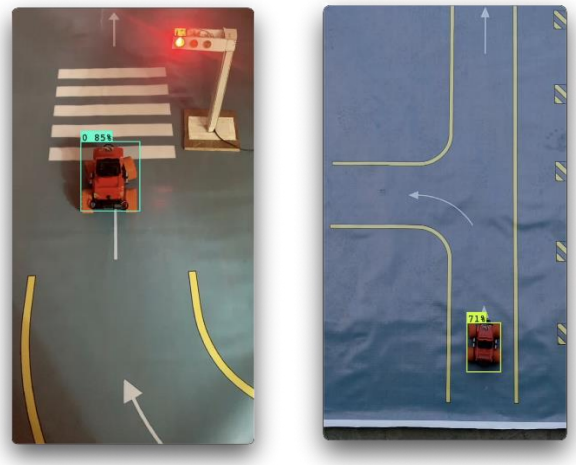


(a) (b) (c) (d)

Figure 10. 8-test: (a) color masking of track, (b) binary masking of track, (c) binary masking of bounding box, (d) car with bounding box prediction on track.



(a) (b)



(c) (d)

Figure 11. Output in different test tracks: (a) Speed Breaker test or the hump test, (b) U-turn test, (c) traffic light violation test, (d) garage parking test.

VI. CONCLUSION

CarSight was trained on the custom dataset and tested on different trial tracks successfully with satisfactory results. Hence the implementation of CarSight can replace the manual driving test system but it has some limitations. To check whether the car is stationary or moving, it counts the number of still frames but, the system sometimes fails or predicts a false positive case if the car is not turned off and stood in the same position for 5 seconds. The system also fails to decide if the car turns off and on within 4 seconds.

VII. ACKNOWLEDGEMENT

We gratefully acknowledge the Pulchowk Campus for its unwavering support and resources.

VIII. REFERENCES

[1] N. Prahith, K. Shridhar, P. Dey and V. Sridhar, “IoT based Automatic Driving License Test,” in International Journal of Engineering Research and Technology, Volume: 9, Issue 4, 2020, pp. 174-177.

[2] S.A. Dodke, “Automation of Driving License test using Wireless Sensor network” in International Research Journal of Engineering and Technology, Volume:2, Issue: 8,2015, pp:1268-1271

[3]

KA. Margale, P. Pawale, A. Patil and J. Waykule, "Driving License Test Automation Using VB," in International Journal of Engineering and Applied Sciences, Volume: 2, Issue: 4, 2015, pp: 15-18.

[4] X. Zhang, Q. Wu, X. Xie, X. Huang, and L. Zhang, "A Driving License Test Monitoring System Based on Computer Vision and Machine Learning" in IEEE Access, vol 9, 2021, pp: 50600-50609.

[5] H. Wang, Q. Li, Y. Song, Q. Liu and Y. Zheng, "Anti-cheating system for driving license tests based on computer vision," in IEEE International Conference on Mechatronics and Automation, 2018, pp. 1323-1328.

[6] J. Redmon, S. Divvala, R. Girshick, and A. Farhadi, "You Only Look Once: Unified, Real-Time Object Detection" in Computer Resource Repository, 2016, pp: 779-788.

[7] AU. Nambi, I. Mehta, A. Ghosh, V. Lingam and V. Padmanabhan, "ALT: Towards Automating Driver License Testing using Smartphones," in Proceedings of the 17th Conference on Embedded Networked Sensor Systems, 2019, pp: 29-42.

[8] S. Cai, G. Ye, J. Li, S. Kuang, Z. Liu, and J. Song, "Kbn-yolov5: Improved yolov5 for detecting bird's nests in high-voltage tower," in China Automation Congress (CAC), 2022, pp: 3020–3025.